Chapter test

37 Pollution control and conservation

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following practices helps alleviate municipal solid waste problem through the concept of 'reduce'?

   A. Use handkerchiefs instead of tissue papers.
   B. Bring our own bags when we go shopping.
   C. Donate used clothes and shoes to charitable organizations.
   D. Use rechargeable batteries instead of alkaline batteries. ☐

2. Which of the following processes helps break down the organic matter in sewage?

   A. screening
   B. sedimentation
   C. pumping air into the sewage
   D. adding chlorine to the sewage ☐

★ 3. Which of the following correctly describes the action of aerobic microorganisms in secondary treatment of the sewage treatment process?

   A. To break down the chemicals added to the sewage in primary treatment.
   B. To break down the organic matter in the sewage.
   C. To break down ammonium compounds in the sewage.
   D. To break down the organic matter in the sludge. ☐

4. Which of the following are the importance of conserving biodiversity to humans?

   (1) It provides a large variety of food sources for humans.
   (2) It provides a large variety of raw materials for humans.
   (3) It provides resources for recreational activities for humans.

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

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5. Which of the following practices help conserve endangered species?

(1) reproducing endangered plant species by artificial propagation
(2) captive breeding and release
(3) setting up regulations to control the sale and possession of endangered species

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

6. Which of the following ***cannot*** help the conservation of the Black-faced Spoonbill?

A. Set up conservation areas.
B. Carry out international census to know more about the number and behaviour of the Black-faced Spoonbill.
C. Modify the genome of Black-faced Spoonbill.
D. Track the positions of the Black-faced Spoonbill by satellite. ☐

★★7. Which of the following statements about *ex situ* conservation are ***incorrect***?

(1) Endangered species are resettled in a man-made, captive environment.
(2) *Ex situ* conservation is especially effective in protecting wild populations of endangered species that are seriously threatened by habitat destruction or hunting.
(3) *Ex situ* conservation provides the most suitable habitat to the endangered species.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

8. Which of the following are measures for sustainable development of fisheries in Hong Kong?

(1) Prohibit destructive fishing practices such as trawling.
(2) Set up artificial reefs in suitable waters.
(3) Rear fish for commercial use.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

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9. Which of the following practices can increase soil fertility in a sustainable manner?

(1) carrying out crop rotation
(2) applying organic fertilizer on soil
(3) using biological pest control

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

★ 10. Which of the following are the advantages of practicing crop rotation?

(1) Less water is used for irrigation.
(2) The pest populations are reduced by interrupting their life cycles.
(3) Soil nutrients can be enriched by growing certain types of crop plants.

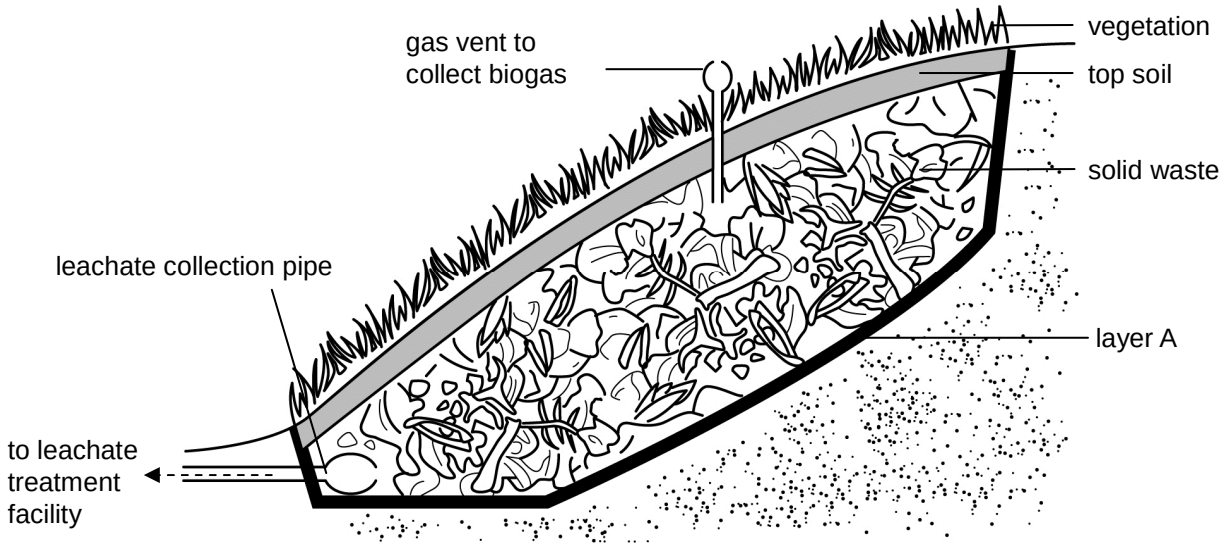
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	37.1	37.1	37.1	37.2	37.3	37.4	37.4	37.4	37.6	37.6

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B. Problem solving (13 marks)

★ 11. The diagram below shows a simplified vertical section of a landfill site. Decomposition of solid wastes produces some gases and liquid. The liquid collected in this way is called leachate.



(a) What is the function of layer A? State ***one*** property that enables it to achieve such function. (3 marks)

(b) The disposal of solid wastes at landfills can be regarded as an example of pollution control and conservation if landfills are managed properly. With reference to the above diagram, suggest how it could control pollution and conserve the environment. (6 marks)

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(c) (i) Recycling is one of the important aspects in solid waste management strategy of Hong Kong. Suggest how recycling can help reduce environmental problems.

(2 marks)

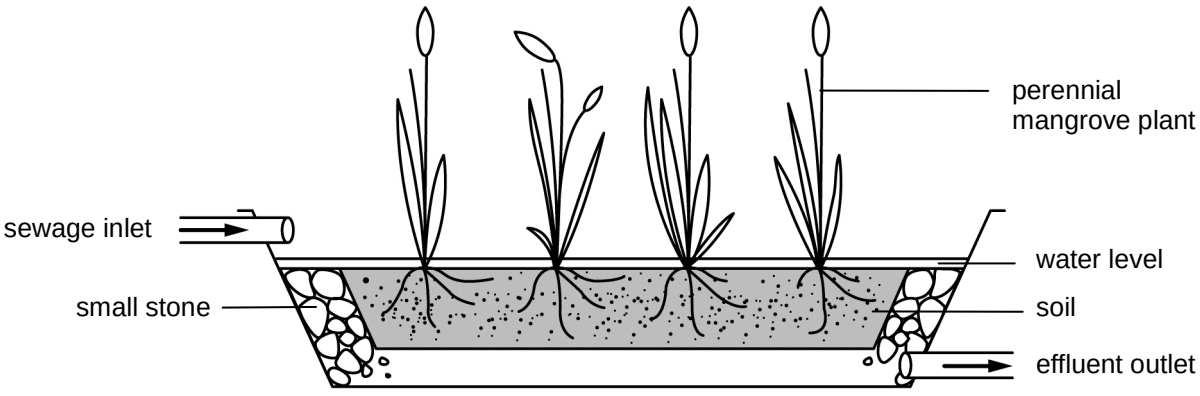
(ii) Suggest ***two*** ways to improve the recycling rate of solid wastes in Hong Kong.

(2 marks)

(Total: 13 marks)

C. Exam challenge (17 marks)

★★12. The diagram below shows an artificial wetland system for sewage treatment. Sewage was used to water some perennial mangrove plant seedlings.



Water samples were collected at the inlet and outlet of the system for analysis. The total nitrogen and phosphorous content, and the number of *Escherichia coli* in the water samples were measured. *E. coil* is commonly used as a biological indicator for water quality.

Results showed that the mean values of various parameters of the sewage at the outlet decreased significantly compared with those at the inlet. The effluent at the outlet was considered safe to be discharged into sea or rivers.

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(a) What are the major pollutants that have to be removed in a domestic sewage treatment works? (3 marks)

(b) According to the diagram, what is / are the stage(s) (primary, secondary or tertiary treatment) involved in this artificial wetland system? Explain your answer. (4 marks)

(c) There are various forms of nitrogen in the sewage. Describe how these forms of nitrogen are made available to the mangrove plants. (3 marks)

(d) What are the unique features of mangrove plants that make them suitable to be used in the artificial wetland system? (3 marks)

(e) Suggest ***two*** advantages of using an artificial wetland system to treat sewage over a sewage treatment works. (2 marks)

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(f) Suggest ***two*** disadvantages of using an artificial wetland system to treat sewage compared to a sewage treatment works. (2 marks)

(Total: 17 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	B	2.	C	3.	B	4.	D	5.	D
6.	C	7.	B	8.	D	9.	A	10.	C

(1 mark each)

B. Problem solving (13 marks)

11. (a) Layer A is used to prevent leaching / leachate from entering the water or soil below, (1)
which would cause contamination of underground water or soil nearby. (1)
Layer A should be impermeable to water / water-proof. (1)

(b) The landfill provides a centralized site for decomposition of solid wastes. (1)
This reduces solid waste problem in other locations. (1)
The leachate collected will be further treated in treatment facility so that it is safe for discharge into sea or rivers. (1)
Some of the gases collected such as methane can be used as fuel (1)
so as to minimize the use of fossil fuels. (1)
The land filled provides space for recreational development after they are stabilized. (1)
(Or other acceptable answers)

(c) (i) Recycling helps reduce the amount of solid wastes disposed of at landfills, reducing the environmental impacts of landfills. (1)
Recycling reduces the amount of energy and raw materials used in the production of new items. (1)
(Or other acceptable answers)

(ii) Set up more waste separation bins. (1)
Implement the charging scheme for municipal solid waste. (1)
(Or other acceptable answers)

(Total: 13 marks)

C. Exam challenge (17 marks)

12. (a) Any ***three*** of the following: (3)
- Large solids (1)
- Organic matter (1)
- Inorganic nutrients (e.g. phosphates) (1)
- Pathogens (1)

(Or other acceptable answers)

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(b) The system involves primary, secondary and tertiary treatments. (1)
Primary treatment: The small stones act as a screen to remove the large solids from the sewage. (1)
Secondary treatment: The microorganisms in the soil break down the organic matter in the sewage. (1)
Tertiary treatment: Inorganic nutrients are removed from the sewage by the mangrove plants. (1)

(c) Organic nitrogenous compounds are broken down to ammonium compounds by aerobic microorganisms in the soil. (1)
Ammonium compounds are converted to nitrates by nitrifying bacteria. (1)
Ammonium compounds / nitrates can be directly absorbed by the mangrove plants. (1)

(d) The mangrove plants are capable to survive in a wide range of soil conditions, such as high salt content. (1)
The mangrove soil and the roots of mangrove plants harbour diverse groups of microorganisms, which may be useful to the breakdown of organic matter in the sewage. (1)
The roots of mangrove plants are capable to survive under anaerobic conditions / in water-logged soil. (1)
(Or other acceptable answers)

(e) Any ***two*** of the following: (2)
- The cost of construction / equipment is lower than that of sewage treatment works. (1)
- The mangrove plants are self-sustaining / no extra manpower needed / lower running cost. (1)
- The artificial wetland helps beautify the scenery. (1)
- The artificial wetland provides habitats to organisms. (1)

(Or other acceptable answers)

(f) Any ***two*** of the following: (2)
- The breakdown of the organic matter in the sewage is slower, as there is no pumping of air into the sewage to facilitate the action of aerobic microorganisms. (1)
- The water quality of the effluent may be inconsistent, depending on the health status of the artificial wetland system. (1)
- The artificial wetland may not be able to treat a very large amount of domestic sewage. (1)

(Or other acceptable answers)

(Total: 17 marks)

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